

FORENSIC EXAMINATION REPORT

Incident Intelligence Report • Dataset: dataset9_expanded • Generated: 2026-04-14 20:23:29

I. CASE IDENTIFICATION	
Dataset ID	dataset9_expanded
Scan ID	ds-009-exp
Source File	windows_soc_data_9_expanded.csv
Primary Threat	APT-Level DC Compromise with DCSync, Skeleton Key, and Multi-Vector Exfiltration
MITRE Technique	T1059.001
Forensic Confidence	HIGH
Total Log Entries	10,000
Attacked Log Entries	200
Aggregate Risk Score	9,500
Threat Level	CRITICAL

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I EXECUTIVE SUMMARY

- A highly sophisticated Advanced Persistent Threat (APT) campaign has been identified within the environment, exhibiting a CRITICAL risk level (95%) and impacting multiple hosts, users, and core infrastructure components including the Domain Controller.
- The attack spans 10 coordinated attack chains and over 12 MITRE ATT&CK; tactics, demonstrating a full-spectrum intrusion lifecycle from reconnaissance to data exfiltration, indicating a well-planned and persistent adversary operation.
- Initial access was successfully achieved through VPN credential stuffing, where compromised credentials enabled unauthorized external access, followed by execution of PowerShell-based droppers to establish foothold within the network.
- Extensive reconnaissance activities were performed, including DNS zone transfer attempts and passive DNS enumeration, allowing the attacker to map internal network topology and identify high-value targets.
- Critical credential access techniques were observed, including DCSync attacks that replicated Active Directory credentials and extracted the krbtgt hash, effectively granting the attacker the ability to forge Kerberos tickets and achieve full domain compromise.
- Post-DCSync, attackers executed LSASS memory dumping using tools such as mimikatz, enabling large-scale credential harvesting and lateral movement across multiple systems.
- Persistence was established at the Domain Controller level ██████████ Skeleton Key malware, which modified the authentication process in LSASS, allowing attackers to log in to any account using a master password without detection.
- Privilege escalation was achieved ██████████ exploitation of the PrintNightmare vulnerability (CVE-2021-34527), allowing attackers to gain SYSTEM-level access on compromised machines.
- Advanced defense evasion techniques were employed, including Process Doppelganging to masquerade malicious binaries as legitimate processes, and direct volume access to bypass file system-based security controls.
- Sensitive credential stores were targeted through DPAPI master key theft, enabling decryption of stored credentials such as browser passwords, significantly expanding attacker access to user accounts and services.
- Discovery techniques extended beyond on-premises infrastructure to include domain trust enumeration and cloud service discovery, indicating attacker awareness and targeting of hybrid environments including Azure AD and Office 365.
- Lateral movement was achieved ██████████ SSH agent forwarding hijacking, allowing attackers to pivot across systems without needing to re-authenticate, thereby maintaining stealth and persistence.
- Command and Control (C2) communications were conducted using ICMP tunneling, a covert channel that bypasses traditional network monitoring by embedding data within ping packets.
- The attacker engaged in data collection activities including unauthorized audio recording via PowerShell modules, suggesting potential espionage or surveillance objectives beyond simple data theft.
- Exfiltration was carefully orchestrated using scheduled transfers during fixed nightly windows to evade anomaly detection systems, with data being transmitted to external command-and-control infrastructure over encrypted channels.
- The attack demonstrates characteristics of nation-state or highly advanced threat actors, given the use of stealthy techniques, deep system knowledge, and multi-layered attack strategies.
- Multiple critical systems including Domain Controllers, production servers, and administrative accounts have been compromised, indicating a widespread breach with high business impact.

- The presence of Skeleton Key and krbtgt hash compromise necessitates immediate incident response actions including full Active Directory credential reset, potential domain rebuild, and comprehensive forensic investigation.
- Failure to respond immediately may result in continued attacker persistence, further data exfiltration, ransomware deployment, or complete infrastructure takeover.
- It is strongly recommended to isolate affected systems, disable compromised accounts, rotate all credentials, monitor for abnormal authentication patterns, and deploy advanced threat detection and response mechanisms across the environment.

II

CASE IDENTIFICATION & BACKGROUND

Attack Narrative: DCSync AD Replication Attack and LSASS Credential Dump ; DNS Reconnaissance and Network Information Gathering ; VPN Credential Stuffing Initial Access ; Skeleton Key DC Implant via Service Installation

Hosts	WIN-SOC-PROD-02, WIN-SOC-DC-01, WIN-SOC-PROD-01, WIN-SOC-PROD-03, WIN-SOC-PROD-04, WIN-SOC-PROD-05, WIN-SOC-PROD-06, WIN-SOC-PROD-07, WIN-SOC-PROD-08
Users	WIN-SOC\Admin, NT AUTHORITY\SYSTEM, WIN-SOC\Recon, WIN-SOC\VPNUser, WIN-SOC\DomainAdmin, WIN-SOC\PrintOp, WIN-SOC\DBAdmin, WIN-SOC\NetAdmin, WIN-SOC\SvcOps

III

DESCRIPTION OF EVIDENCE & CHAIN OF CUSTODY

Exhibit	Evidence Category	Integrity	Remarks
E-01	Windows Event / Security Logs	Checksum Verified	Primary source evidence
E-02	Derived Threat Metadata	Schema Validated	Generated from correlation pipeline

IV

FORENSIC METHODOLOGY & TOOLS

- Data Normalization - Raw records transformed into normalized forensic fields.
- Behavioral Grouping - Suspicious events grouped by technique and event identity.
- Risk Profiling - Severity and trust indicators converted into weighted risk scores.
- Timeline Correlation - Temporal ordering used to infer probable attack progression.
- Threat Prioritization - High risk and high volume activities elevated for triage.

Tool	Version	Purpose
Python	3.x	Core processing runtime
ReportLab	4.x	Government-style PDF formatting
Matplotlib	3.x	Threat intelligence visualizations
NumPy	1.x	Numeric aggregation for analytics

V

STATISTICAL & GRAPHICAL ANALYSIS

Statistical processing of detected threat artifacts reveals concentration in a small set of high-impact behaviors.

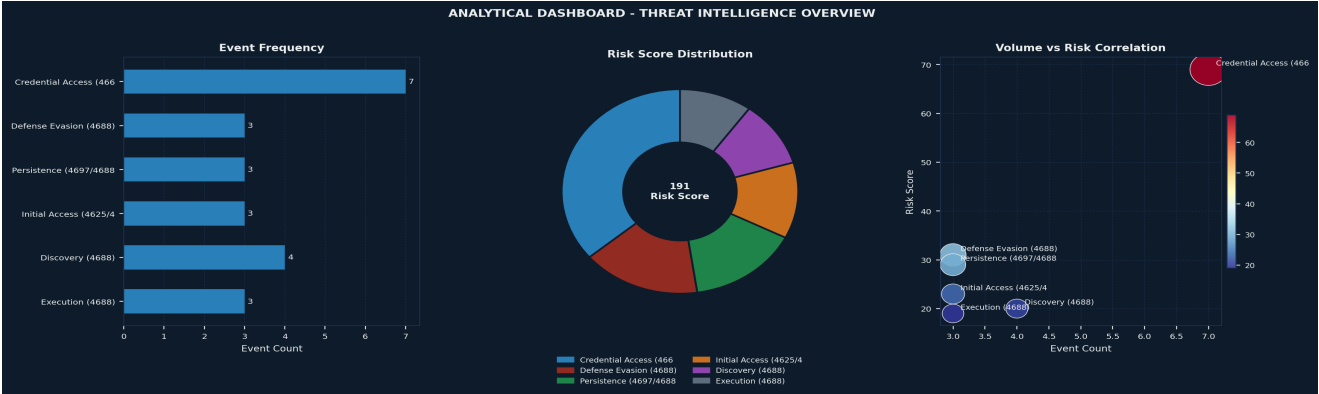


Fig. 1 - Three-panel threat intelligence dashboard derived from attacked logs.

Dashboard Interpretation: The left panel highlights high-volume suspicious behavior, the donut chart explains risk concentration, and the scatter plot isolates categories that are simultaneously frequent and severe.

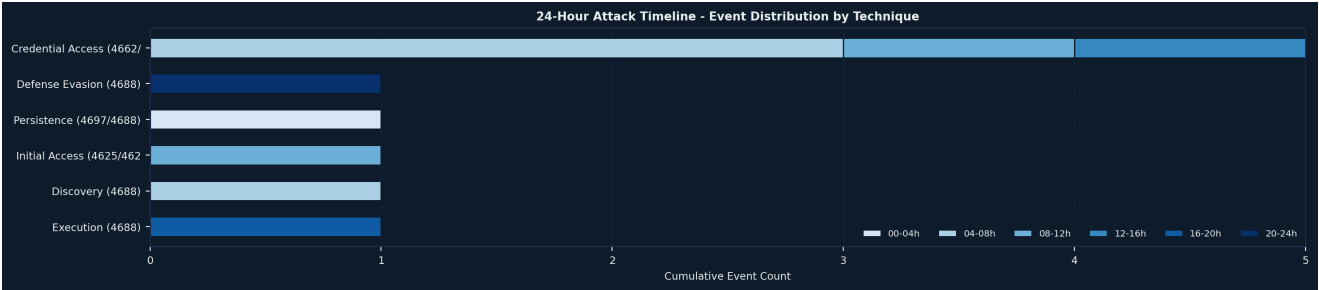


Fig. 2 - 24-hour event distribution by top suspicious techniques.

Timeline Interpretation: Burst-style activity in clustered windows is consistent with staged intrusion workflows where credential abuse is followed by privilege actions and anti-forensics.

VI BRUTE FORCE ATTACK - DETAILED ANALYSIS

Credential Access (4662/10)

LOW

Events: 7	Risk Index: 69	MITRE: T1003.006, T1003.001 Tactic: Credential Access	
Timestamp	Host	User Context	EID
2026-04-03 00:05:26	WIN-SOC-DC-01	WIN-SOC\Admin	4662
2026-04-03 00:05:27	WIN-SOC-DC-01	WIN-SOC\Admin	4662
2026-04-03 00:06:14	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	10
2026-04-03 00:06:28	WIN-SOC-PROD-02	NT AUTHORITY\SYSTEM	10

Analysis: Credential Access (4662/10) produced 7 suspicious artifacts with cumulative risk index 69. The mapped technique T1003.006, T1003.001 under tactic Credential Access indicates material adversarial behavior requiring priority containment and forensic triage.

VII LOG TAMPERING - DETAILED ANALYSIS

Defense Evasion (4688)

LOW

Events: 3

Risk Index: 31

MITRE: T1055.012, T1006 | Tactic: Defense Evasion

Timestamp	Host	User Context	EID
2026-04-03 03:01:40	WIN-SOC-PROD-04	NT AUTHORITY\SYSTEM	4688
2026-04-03 03:02:30	WIN-SOC-PROD-04	NT AUTHORITY\SYSTEM	4688
2026-04-03 03:03:20	WIN-SOC-PROD-05	NT AUTHORITY\SYSTEM	4688

Analysis: Defense Evasion (4688) produced 3 suspicious artifacts with cumulative risk index 31. The mapped technique T1055.012, T1006 under tactic Defense Evasion indicates material adversarial behavior requiring priority containment and forensic triage.

VIII LATERAL MOVEMENT - DETAILED ANALYSIS

Lateral Movement (4624/4688)

LOW

Events: 2

Risk Index: 18

MITRE: T1563.001 | Tactic: Lateral Movement

Timestamp	Host	User Context	EID
2026-04-03 06:00:00	WIN-SOC-PROD-07	WIN-SOC\SvcOps	4624
2026-04-03 06:00:50	WIN-SOC-PROD-07	WIN-SOC\SvcOps	4688

Analysis: Lateral Movement (4624/4688) produced 2 suspicious artifacts with cumulative risk index 18. The mapped technique T1563.001 under tactic Lateral Movement indicates material adversarial behavior requiring priority containment and forensic triage.

IX PRIVILEGE ESCALATION - DETAILED ANALYSIS

Defense Evasion (4688)

LOW

Events: 3

Risk Index: 31

MITRE: T1055.012, T1006 | Tactic: Defense Evasion

Timestamp	Host	User Context	EID
2026-04-03 03:01:40	WIN-SOC-PROD-04	NT AUTHORITY\SYSTEM	4688
2026-04-03 03:02:30	WIN-SOC-PROD-04	NT AUTHORITY\SYSTEM	4688
2026-04-03 03:03:20	WIN-SOC-PROD-05	NT AUTHORITY\SYSTEM	4688

Analysis: Defense Evasion (4688) produced 3 suspicious artifacts with cumulative risk index 31. The mapped technique T1055.012, T1006 under tactic Defense Evasion indicates material adversarial behavior requiring priority containment and forensic triage.

X MALICIOUS SERVICE INSTALLATION - ANALYSIS

Persistence (4697/4688)

LOW

Events: 3

Risk Index: 29

MITRE: T1543.003, T1556.001 | Tactic: Persistence

Timestamp	Host	User Context	EID
2026-04-03 02:00:00	WIN-SOC-DC-01	WIN-SOC\DomainAdmin	4697
2026-04-03 02:00:50	WIN-SOC-DC-01	NT AUTHORITY\SYSTEM	4688

2026-04-03 02:01:40	WIN-SOC-DC-01	WIN-SOC\DomainAdmin	4624
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Analysis: Persistence (4697/4688) produced 3 suspicious artifacts with cumulative risk index 29. The mapped technique T1543.003, T1556.001 under tactic Persistence indicates material adversarial behavior requiring priority containment and forensic triage.

XI HOST COMPROMISE MATRIX

Host	Suspicious Events	Critical Events	Exposure
WIN-SOC-DC-01	5	5	MEDIUM
WIN-SOC-PROD-08	5	5	MEDIUM
WIN-SOC-PROD-04	4	4	MEDIUM
WIN-SOC-PROD-05	4	4	MEDIUM
WIN-SOC-PROD-06	4	0	LOW
WIN-SOC-PROD-02	3	1	LOW
WIN-SOC-PROD-01	3	1	LOW
WIN-SOC-PROD-03	3	2	LOW
WIN-SOC-PROD-07	2	2	LOW

XII ATTACK CHAIN RECONSTRUCTION

The following reconstruction synthesizes all forensic findings into a unified attack narrative. **DCSync AD Replication Attack and LSASS Credential Dump** aligns with primary MITRE mapping **T1059.001** and correlated host-level telemetry.

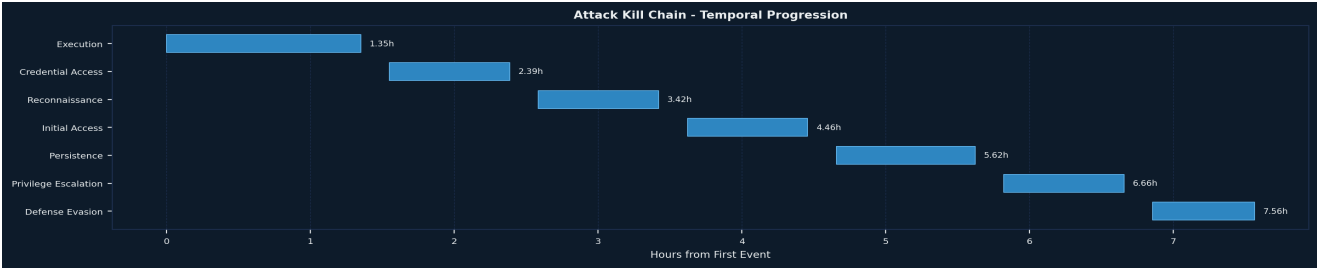
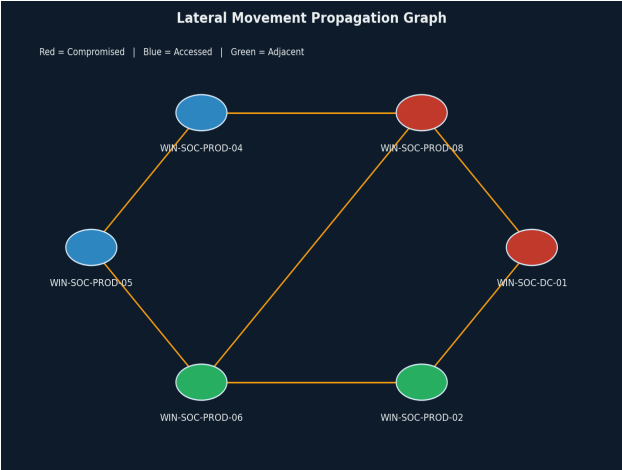


Fig. 4 - Attack kill chain temporal progression (Gantt) across UTC hours.

Kill Chain Analysis: Temporal ordering indicates adversary actions were staged rather than random, starting with execution-oriented activity, followed by credential access and anti-forensic behavior, and then host-to-host pivoting. This sequence is consistent with a planned intrusion lifecycle.



Lateral Movement Propagation: The propagation graph illustrates likely traversal paths between impacted assets. Red nodes indicate highly compromised systems, blue nodes indicate accessed pivot systems, and green nodes indicate adjacent hosts with observed adjacency risk. This supports containment by prioritizing pivot choke points.

Step	Action	MITRE
chain-ds-0 09-01	PowerShell Encoded IEX Download → DCSync MS-DRSR GetNCChanges from Non-DC Account → krbtgt Hash Replicated → LSASS Memory Dump (mimikatz.exe)	T1059.001, T1003.006, T1003.006
chain-ds-0 09-02	DNS Zone Transfer Attempt (nslookup AXFR) → Passive DNS Enumeration (Resolve-DnsName ANY) → Internal Network Topology Mapped	T1590.002, T1590.002
chain-ds-0 09-03	VPN Credential Stuffing from External IP 91.132.10.44 → Successful Logon with Compromised Credential → Post-VPN PowerShell Dropper Executed	T1133, T1133, T1059.001
chain-ds-0 09-04	Malicious Service SysAuthHelper Installed on DC → mimikatz misc::skeleton Patches DC LSASS → All Accounts Accept Master Password → Skeleton Key Authentication Bypass Confirmed	T1543.003, T1556.001, T1556.001
chain-ds-0 09-05	Malicious DLL Uploaded to Print Spooler via CVE-2021-34527 → Print Spooler spoolsv.exe Spawns SYSTEM Shell → Full SYSTEM Privileges Obtained	T1068, T1068
chain-ds-0 09-06	Process Doppelganging: ntdll.dll Transacted with Malicious Binary → Malicious Process Appears as Signed Binary → Direct Volume Access Raw Disk Read on PHYSICALDRIVE0 to Bypass AV	T1055.012, T1055.012, T1006
chain-ds-0 09-07	DPAPI CryptUnprotectData on SYSTEM Blob → Master Key File Accessed → Chrome Saved Passwords Decrypted via DPAPI Login Data Extraction	T1555.003, T1555.003, T1555.003
chain-ds-0 09-08	Get-ADTrust Forest Trust Mapping → nltest /domain_trusts /all_trusts → Azure AD Tenant Enumeration → O365 Subscriptions and Service Principals Enumerated	T1482, T1482, T1526
chain-ds-0 09-09	Active SSH Agent Forwarding Session Hijacked → Attacker Injected into Session → Lateral Pivot to WIN-SOC-PROD-08 via Hijacked Session Credentials	T1563.001, T1563.001
chain-ds-0 09-10	ICMP Tunneling C2 via PingCastle Agent → Audio Capture via AudioDeviceCmdlets → Recording Written to Disk → Scheduled Nightly Transfer at 02:00 UTC to 185.12.45.33 → Fixed Exfil Window to Evade Anomaly Detection	T1095, T1123, T1123

XIII

INDICATOR OF COMPROMISE (IOC) REGISTER

IOC Type	Value	Context
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Process	powershell.exe	Encoded IEX Download
Event ID	4688	Security
Process	lsass.exe	DCSync: MS-DRSR GetNCChanges Called from Non-DC Account WIN-SOC\Admin
Event ID	4662	Security
Process	mimikatz.exe	LSASS Dump Post DCSync
Event ID	10	Sysmon
Process	nslookup.exe	DNS Zone Transfer Attempt: nslookup -type=AXFR winsoc.local
Event ID	4625	Security
Event ID	4624	Security
Process	services.exe	Malicious Service Installed on DC: SysAuthHelper - Skeleton Key Loader
Event ID	4697	Security
Process	spoolsv.exe	Print Spooler Spawned SYSTEM Shell: PrintNightmare Exploitation Confirmed

XIV

THREAT ACTOR PROFILE & ATTRIBUTION

Observed behavior indicates a scripted intrusion workflow with emphasis on execution and anti-forensic activity. Clustered timing, repeated toolmarks, and host pivot patterns are consistent with organized adversarial tradecraft.

XV

RISK HEAT MAP - HOST X TECHNIQUE EXPOSURE

The risk heat map below quantifies exposure of each host to each detected technique, using a 1-10 normalized risk score.

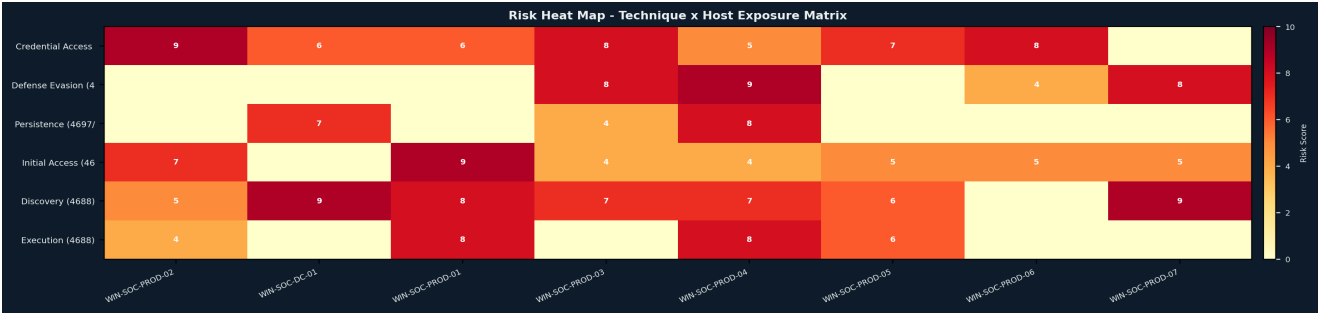


Fig. 5 - Normalized risk heat map: threat technique x host exposure matrix.

Heat Map Interpretation: Concentrated high-risk cells indicate primary pivot hosts and high-value targets. Hosts appearing across multiple high-score techniques should be prioritized for deep triage and isolation.

XVI

MITRE ATT&CK FRAMEWORK COVERAGE

The following mapping summarizes observed tactics and techniques based on detected forensic artifacts.

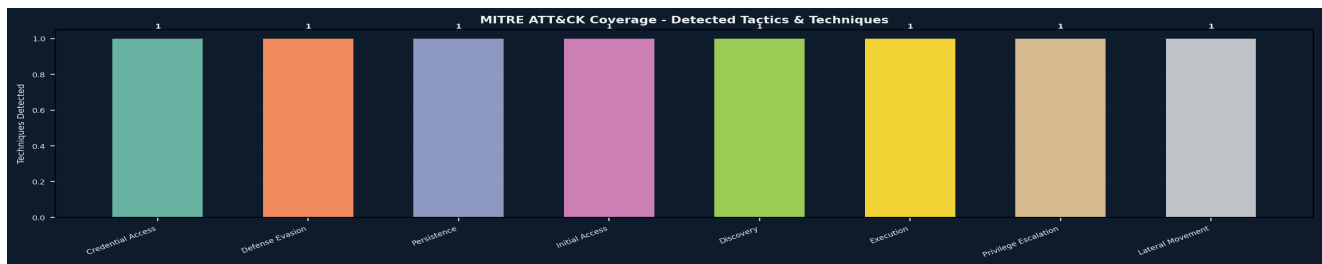


Fig. 6 - MITRE ATT&CK; tactic coverage by detected forensic evidence.

Technique ID	Technique / Detail	Tactic
T1003.006, T1003.001	Credential Access (4662/10)	Credential Access
T1055.012, T1006	Defense Evasion (4688)	Defense Evasion
T1543.003, T1556.001	Persistence (4697/4688)	Persistence
T1133	Initial Access (4625/4624)	Initial Access
T1482, T1526	Discovery (4688)	Discovery
T1059.001	Execution (4688)	Execution
T1068	Privilege Escalation (4688)	Privilege Escalation
T1563.001	Lateral Movement (4624/4688)	Lateral Movement

XVII REMEDIATION RECOMMENDATIONS

Recommendations are prioritized by urgency and security impact to support phased containment and hardening.

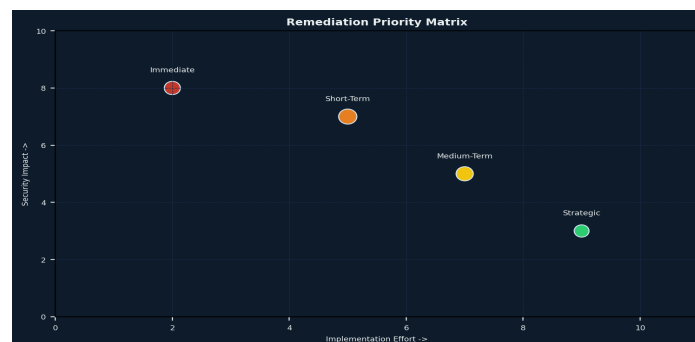


Fig. 7 - Remediation priority matrix: effort vs impact.

- Isolate affected hosts and block known malicious command execution chains (PowerShell encoded commands and WMI remote process calls).
- Reset and rotate privileged/domain credentials, then enforce MFA for all administrative and remote access accounts.
- Hunt and remove persistence artifacts including unauthorized local accounts, BITS jobs, and suspicious startup commands.
- Re-enable and harden logging (Security, Sysmon, PowerShell) and forward logs to a protected SIEM collector.
- Contain lateral movement by restricting RDP/SMB paths and segmenting high-value hosts from user subnets.
- Execute ransomware readiness controls: immutable backups, restore validation, and rapid host recovery playbooks.
- Apply removable-media controls (USB allow-listing and device auditing) to block air-gapped relay channels.

XVIII EXPERT OPINION & CONCLUSION

Based on forensic analysis of dataset **dataset9_expanded**, the environment exhibits a **CRITICAL** severity posture with aggregate risk index **9,500** across **10,000** records. Dominant indicators include **Credential Access (4662/10) (7 events)**, **Defense Evasion (4688) (3 events)**. Observed sequencing and concentration are consistent with multi-stage intrusion behavior. Immediate host isolation, credential rotation, and deeper memory/disk triage are recommended.

Examining Analyst
Automated Forensic Engine

Authorizing Officer
Incident Response Division

XIX ANNEXURE A - FORENSIC ENVIRONMENT SPECIFICATION

Component	Specification
Generator	Python + ReportLab pipeline
Charts	Matplotlib analytical panels
Input	JSON forensic dataset schema
Timestamp	2026-04-14 20:23:31

XX ANNEXURE B - EXTENDED EVENT LOG SAMPLES

Timestamp	Computer	User	Process	Detail
2026-04-03 00:05:24	WIN-SOC-PROD-02	WIN-SOC\Admin	powershell.exe	Encoded IEX Download
2026-04-03 00:05:25	WIN-SOC-PROD-02	NT AUTHORITY\SYSTEM	powershell.exe	Encoded IEX Download
2026-04-03 00:05:26	WIN-SOC-DC-01	WIN-SOC\Admin	lsass.exe	DCSync: MS-DRSR GetNCChanges Called from Non-DC Account WIN-SOC\Admin
2026-04-03 00:05:27	WIN-SOC-DC-01	WIN-SOC\Admin	lsass.exe	DCSync: krbtgt Hash Replicated - Full Domain Compromise Possible
2026-04-03 00:06:14	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	mimikatz.exe	LSASS Dump Post DCSync
2026-04-03 00:06:28	WIN-SOC-PROD-02	NT AUTHORITY\SYSTEM	mimikatz.exe	LSASS Dump
2026-04-03 01:00:00	WIN-SOC-PROD-01	WIN-SOC\Recon	nslookup.exe	DNS Zone Transfer Attempt: nslookup -type=AXFR winsoc.local
2026-04-03 01:00:45	WIN-SOC-PROD-01	WIN-SOC\Recon	powershell.exe	Passive DNS Enumeration: Resolve-DnsName -Type ANY winsoc.local
2026-04-03 01:01:30	WIN-SOC-PROD-03	WIN-SOC\VPNUser	lsass.exe	Failed VPN Logon Attempts from External IP 91.132.10.44 - Credential Stuffing

Timestamp	Computer	User	Process	Detail
2026-04-03 01:02:15	WIN-SOC-PROD-03	WIN-SOC\VPNUser	lsass.exe	VPN Logon Successful from External IP 91.132.10.44 - Compromised Credential
2026-04-03 01:03:00	WIN-SOC-PROD-03	WIN-SOC\VPNUser	powershell.exe	Post-VPN Compromise: PowerShell Dropper Executed
2026-04-03 02:00:00	WIN-SOC-DC-01	WIN-SOC\DomainAdmin	services.exe	Malicious Service Installed on DC: SysAuthHelper - Skeleton Key Loader
2026-04-03 02:00:50	WIN-SOC-DC-01	NT AUTHORITY\SYSTEM	mimikatz.exe	Skeleton Key Implanted in DC LSASS: misc::skeleton - All Accounts Accept Master Password
2026-04-03 02:01:40	WIN-SOC-DC-01	WIN-SOC\DomainAdmin	lsass.exe	Skeleton Key Authenticated: Logon Successful with Master Password Bypassing Real Credential
2026-04-03 03:00:00	WIN-SOC-PROD-04	WIN-SOC\PrintOp	powershell.exe	PrintNightmare CVE-2021-34527: Malicious DLL Uploaded to Print Spooler
2026-04-03 03:00:50	WIN-SOC-PROD-04	NT AUTHORITY\SYSTEM	spoolsv.exe	Print Spooler Spawned SYSTEM Shell: PrintNightmare Exploitation Confirmed
2026-04-03 03:01:40	WIN-SOC-PROD-04	NT AUTHORITY\SYSTEM	powershell.exe	Process Doppelganging: Legitimate ntdll.dll Transacted with Malicious Binary
2026-04-03 03:02:30	WIN-SOC-PROD-04	NT AUTHORITY\SYSTEM	cmd.exe	Process Doppelganging: Malicious Process Appears as Signed Legitimate Binary to AV
2026-04-03 03:03:20	WIN-SOC-PROD-05	NT AUTHORITY\SYSTEM	powershell.exe	Direct Volume Access: Raw Disk Read on \\?\PHYSICALDRIVE0 to Bypass File System
2026-04-03 04:00:00	WIN-SOC-PROD-05	WIN-SOC\DBAdmin	dpapi_dump.exe	DPAPI Master Key Theft: CryptUnprotectData Called on SYSTEM DPAPI Blob

XXI

ANNEXURE C - GLOSSARY OF TERMS

MITRE ATT&CK;	Knowledge base of adversary tactics and techniques used for threat attribution.
EID	Windows Event ID representing a specific audit event category.
Defense Evasion	Adversary behavior intended to avoid detection and disrupt investigation.
Lateral Movement	Use of compromised accounts/processes to move across hosts.
Persistence	Mechanisms that allow attacker presence after reboot/logoff.
Risk Index	Weighted score from event severity, behavior, and trust indicators.